

PUBLIC VERSION

**UNITED STATES INTERNATIONAL TRADE COMMISSION
WASHINGTON, D.C.**

**Before the Honorable Clark S. Cheney
Chief Administrative Law Judge**

In the Matter of

**CERTAIN SEMICONDUCTOR
DEVICES, PRODUCTS CONTAINING
SAME, AND COMPONENTS THEREOF**

Investigation No. 337-TA-1500

RESPONDENTS' RESPONSIVE BRIEF ON CLAIM CONSTRUCTION

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EXHIBIT LIST

Exhibit	Description
A	Declaration of Dr. Sanjay K. Banerjee Regarding Claim Construction
B	Declaration of Dr. Frank Ernst Regarding Claim Construction
C	Declaration of Dr. David K. Choi Regarding Claim Construction
D	Dr. Sanjay K. Banerjee Curriculum Vitae
E	Dr. Frank Ernst Curriculum Vitae
F	Dr. David K. Choi Curriculum Vitae
G	Responsive Declaration of Dr. Sanjay K. Banerjee Regarding Claim Construction
1	Complainant's Infringement Contentions, Ex. A4
2	<i>Refractory Metal</i> , ScienceDirect, https://www.sciencedirect.com/topics/materials-science/refractory-metal (last visited August 24, 2026)
3	<i>Titanium</i> , Royal Society of Chemistry, https://periodic-table.rsc.org/element/22/titanium (last visited August 24, 2026)
4	U.S. Patent Application Publication No. 2012/0161324 A1 ("Heinrich")
5	U.S. Patent Application Publication No. 2014/0027883 A1 ("Hu")
6	U.S. Patent Application Publication No. 2015/0221740 A1 ("Alptekin")
7	<i>Intellectual Ventures I LLC v. Canon Inc.</i> , C.A. No. 11-792-SLR, Memo. Ord. re Claim Constr. (D. Del. Apr. 10, 2014), Dkt. No. 252
8	<i>Bridgelux, Inc. v. Cree, Inc.</i> , C.A. No. 06-6495-PJH, Ord. Construing Claims (N.D. Cal. Aug. 15, 2008), Dkt. No. 109
9	<i>Advanced Tech. Incubator, Inc. v. Sharp Corporation</i> , C.A. No. 2:07-cv-468-BAB, Claim Constr. Ord. (E.D. Tex. Mar. 11, 2009), Dkt. No. 117
10	Complainant's Technical Domestic Industry Contentions, Ex. B6
11	Technical document (GF-ITC-00006842)
12	Merriam-Webster's Collegiate Dictionary (11th Ed.), 2003 (definition of "intermix")
13	<i>Excerpt</i> of Prosecution History of the U.S. Patent No. 9,865,546 (GF-ITC-00004771 – GF-ITC-00005023)
14	<i>Excerpt</i> of Prosecution History of the U.S. Patent No. 10,707,167 (GF-ITC-00004128 – GF-ITC-00004309)
15	Chow, et al., <i>Silicide Formation from Laser Thermal Processing of Ti/Co Bilayers</i> (2004)
16	Email between counsel for Complainant and Respondents re claim construction
17	Praneet Adusumilli et al, <i>Silicide-phase evolution and platinum redistribution during silicidation of Ni_{0.95}Pt_{0.05}/Si(100) specimens</i> , 112 J. Appl. Phys. 064307 (2012) ("Adusumilli")

EXHIBIT LIST (cont.)

Exhibit	Description
18	'235 patent <i>Excerpt</i> from Complainant GlobalFoundries U.S. Inc.'s Second Supplemental Objections and Responses to Respondents' First Set of Interrogatories (Nos. 1-73) ("GF validity contentions")
19	Complainant's Infringement Contentions, Ex. A1
20	Complainant's Domestic Industry Contentions, Ex. B1
21	U.S. Patent Application Publication No. 2008/0191291A1 ("Wu '291") (TOWER_1500_00034458)
22	U.S. Patent Application Publication No. 2009/0020811A1 ("Voldman '811") (TOWER_1500_00034410)
23	This Exhibit Was Intentionally Left Blank
24	'983 patent prosecution history: July 16, 2012 Application (GF-ITC-00004497 – 4571)
25	U.S. Patent No. 6,979,864 ("Negoro")
26	U.S. Patent No. 7,935,992 ("Chang")
27	'983 patent prosecution history: Jan. 25, 2013 Amendment (GF-ITC-00004596 – 4610)
28	'983 patent prosecution history: October 26, 2012 Non-Final Rejection (GF-ITC-00004576 – 4593)
29	U.S. Patent No. 8,053,319 ("Liu")
30	U.S. Patent No. 8,222,130 ("Zhang")
31	<i>Excerpt</i> from <i>Micro-Electronic Circuits</i> , Adel S. Sedra et al., CBS College Publishing (1982) (TOWER_1500_00040579)
32	'983 patent <i>Excerpt</i> from Complainant GlobalFoundries U.S. Inc.'s Second Supplemental Objections and Responses to Respondents' First Set of Interrogatories (Nos. 1-73) ("GF validity contentions")
33	<i>Excerpt</i> from <i>Micro-Electronic Circuits</i> , Adel S. Sedra et al., Saunders College Publishing (1991)
34	Ashwin Shah et al., <i>A 4Mb DRAM with cross point trench transistor cell</i> , SC-21:5 IEEE J. ISSCC Dig. Tech. Papers, 618-626 (1986)
35	Shaw Banerjee & D. Mark Bordelon, <i>A Model for the Trench Transistor</i> , ED-34:12 IEEE Transactions on Electron Devices 2485, 2485-2492 (1987)
36	Shaw Banerjee et al, <i>Leakage Mechanisms in the Trench Transistor DRAM Cell</i> , ED-35:1 IEEE Transactions on Electron Devices, 108-116 (1988)
37	<i>Matsushita Elec. Indus. Co. v. Samsung Elecs. Co.</i> , C.A. No. 3:02-cv-00336-GEB-MCA, Dkt. No. 107 (D.N.J. Mar. 10, 2006)
38	<i>Lexington Luminance LLC v. Feit Elec. Co.</i> , C.A. No. 18-10513 PSG (KSx), Dkt. No. 124 (C.D. Cal. Jun. 24, 2020)

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40	<i>Lexington Luminance LLC v. Feit Elec. Co.</i> , C.A. No. 12-11554-WGY, Dkt. No. 43 (D. Mass. Jun. 25, 2013)
41	<i>Tr. of Boston Univ. v. Everlight Elecs. Co.</i> , C.A. No. 12-11935-PBS, Dkt. No. 511 (D. Mass. May 20, 2014)
42	<i>Lexington Luminance LLC v. Amazon.com, Inc.</i> , C.A. No. 12-12216-DJC, Dkt. No. 196 (D. Mass. Apr. 4, 2016)
43	<i>Lexington Luminance LLC v. Google, Inc.</i> , C.A. No. 1:12-cv-12218-RGS, Dkt. No. 113 (D. Mass. Mar. 29, 2016)
44	<i>Seoul Semiconductor Co. v. Nichia Corp.</i> , C.A. No. 9:07-cv-273, Dkt. No. 86 (E.D. Tex. Feb. 6, 2009)
45	<i>Certain Liquid Crystal Display Devices, Including Monitors, Televisions, & Modules, & Components Thereof</i> , Inv. No. 337-TA-749, Comm’n Op., EDIS No. 484787 (Jul. 6, 2012)
46	August 20, 2026 Email between counsel for Respondents and Complainant re claim construction
47	Prosecution History of the U.S. Patent No. 8,330,235 (GF-ITC-00004310 – GF-ITC-00004484)

I. INTRODUCTION

For the terms amenable to construction, Respondents have proposed constructions that are grounded in the language of the claims, align with the intrinsic record, and properly resolve disputes over claim scope. Respondents' constructions are thus both necessary and proper. *O2 Micro Int'l, Ltd. v. Beyond Innovation Tech. Co.*, 521 F.3d 1351, 1361 (Fed. Cir. 2008).

Respondents further identified certain terms that are indefinite because the specification and prosecution history fail to provide sufficient guidance to delineate the scope of the claims. Complainant does not identify any objective boundaries in the patents or the specifications that define the scope of these terms. Instead, under the guise of plain-and-ordinary meaning, Complainant has embraced the ambiguity of these terms so that it can arbitrarily argue that the accused products fall within the scope but the prior art does not. This is not proper. Because the patents, including their specifications and file histories, fail to provide the reasonable certainty required by the law, these terms should be found indefinite. *Nautilus, Inc. v. Biosig Instruments, Inc.*, 572 U.S. 898 (2014).

II. ARGUMENT

A. “layer” / “first liner layer” / “refractory metal layer”

Disputed Claim Term	Complainant's Proposed Construction	Respondents' Proposed Construction
“layer” / “first liner layer” / “refractory metal layer” ('546 patent, claims 1-14; '167 patent, claims 1, 3, and 4)	Plain and ordinary meaning; not indefinite.	Indefinite or, if not indefinite: “a separate and defined [first liner / refractory metal] coating of material”

Complainant asserts that the term “‘layer’ is one of the most basic and widely used words in the semiconductor art.” Complainant Open. Br. at 6. However, even accepting this statement as true, it does not support adoption of Complainant's plain-and-ordinary meaning construction.

A search of patent cases involving semiconductor-related technologies shows that a large majority of tribunals have **not** adopted a plain-and-ordinary meaning construction without elaboration for the term “layer.”¹ See Ex. 9, *Advanced Tech. Incubator, Inc. v. Sharp Corp.*, C.A. No. 2:07-cv-468-BAB, Dkt. No. 117 (E.D. Tex. Mar. 11, 2009), at 17 (construing “layer” to mean a “separate layer”); Ex. 8, *Bridgelux, Inc. v. Cree, Inc.*, C.A. No. C 06-6495-PJH, Dkt. No. 109 (N.D. Cal. Aug. 15, 2008), at 11 (construing “layer” to mean “[a] defined thickness which is part of a material”); Ex. 37, *Matsushita Elec. Indus. Co. v. Samsung Elecs. Co.*, C.A. No. 3:02-cv-00336-GEB-MCA, Dkt. No. 107 (D.N.J. Mar. 10, 2006), at 61 (“‘layer’ means ‘a thickness of material laid on, spread over, or otherwise covering a surface; a stratum’”); see also Ex. 38 (*Lexington Luminance LLC v. Feit Elec. Co.*) at 10; Ex. 39 (*LG Display Co. v. Au Optronics Corp.*) at 16–17; Ex. 7 (*Intell. Ventures I LLC v. Canon Inc.*) at 9; Ex. 40 (*Lexington Luminance LLC v. Feit Elec. Co.*) at 3; Ex. 41 (*Tr. of Boston Univ. v. Everlight Elecs. Co.*) at 30; Ex. 42 (*Lexington Luminance LLC v. Amazon.com, Inc.*) at 10; Ex. 43 (*Lexington Luminance LLC v. Google, Inc.*) at 11; Ex. 44 (*Seoul Semiconductor Co. v. Nichia Corp.*) at 15; Ex. 45 (Inv. No. 337-TA-749) at 10–12.

In this case, like in many of those cases, construction is necessary to address two claim scope disputes that, without further guidance, would otherwise render the claims indefinite. Ex. 9, *Advanced Tech.*, C.A. No. 2:7-cv-00468, Claim Constr. Ord. at 17 (ruling that, “if a ‘layer’ were not considered ‘separate and distinct,’ then the Parties could argue at trial that the term ‘layer’ be interpreted as several layers which are co-mingled and which are not considered either separate or distinct” and that “[s]uch a construction would contradict the ordinary meaning of the term ‘layer’”).

¹ A search for decisions construing the term “layer” using Docket Navigator yields 86 results, 17 of which relate to semiconductors or adjacent technologies. Only five of these 17 results adopted a plain-and-ordinary meaning construction without elaboration.

First, Respondents’ construction clarifies that a single coating of material cannot be subdivided into multiple layers—i.e., a layer must be a “*separate* and defined . . . coating of material.” Neither Complainant nor its expert provides any explanation regarding when or why a POSA can subdivide a single coating of material into two (or more) different “layers.” Banerjee Responsive Decl., ¶ 7; Banerjee Decl., ¶¶ 56-59. Instead, Complainant’s sole dispute with the requirement that a layer must be a “separate . . . coating of material” is that it purportedly excludes the embodiment of a first liner layer that partially diffuses into an intermix region. Complainant Open. Br. at 8. Not so. There is no dispute that the first liner layer is a separate and defined coating of material in both the ’546 and ’167 patents’ claims. Banerjee Responsive Decl., ¶ 12. The fact that a subsequent method step recites creation of a “first intermix *region*” (not a “layer”) that includes a portion of the first liner layer does not change the fact that a POSA would have understood the first liner layer, as originally deposited, to be a separate and defined coating of material. *Id.*

Second, Respondents’ construction clarifies that multiple coatings of material cannot be arbitrarily combined and subdivided into layers—a layer must be a “separate *and defined* . . . coating of material.” Complainant contends that this aspect of Respondents’ construction is improper because the patents’ specification teaches the “dielectric layer . . . may include . . . more than two separate layers” and the “refractory metal layer” may be a “bilayer.” Complainant Open. Br. at 7 (citing ’546 patent, 4:56-62, 7:45-46). But Respondents’ alternative proposed construction is consistent with these embodiments and does not require a coating of material that is uniformly made up of only a single element or alloy. Banerjee Responsive Decl., ¶ 11. Instead, the material could comprise a bilayer of Ti/TiN for refractory metal purposes or a bilayer of different dielectric materials, so long as the larger “layer” is identifiable by its purpose such that it is separate and

defined. *See* '546 patent, 4:54-5:23 (describing bilayer of different dielectric materials, which the POSA would recognize as being a “dielectric layer” because of the collective purpose of such layers to serve as an electrically insulating material); *id.*, 6:64-7:23 (describing bilayer of Ti/TiN that would have been known to serve adhesion (Ti) and barrier (TiN) purposes in the contact stack); Banerjee Responsive Decl., ¶ 11. In other words, the “material” referenced in Respondents’ alternative proposed construction need not be a uniform thickness of a single element or alloy. *Id.* It could be made up of different elements or alloys, so long as the “material” of the layer is separate and defined. *Id.* Complainant does not explain how a POSA could understand the scope of the claims with reasonable certainty if layers are not separate and defined.

Complainant’s remaining arguments do not avoid the need for a clarifying construction.

First, Complainant cites Respondent’s webpage that uses the term “layer” to suggest that “layer” needs no construction. Complainant Open. Br. at 6 (citing Ex. 6). But the content of this exhibit refers to separate and defined coatings of materials, such as Cu or Al, consistent with the plain-and-ordinary meaning of the term “layer” as reflected in Respondents’ proposed construction. Banerjee Responsive Decl., ¶ 8.

Second, it is not enough, as Complainant contends, that the '546 and '167 patents’ claims recite certain limitations regarding the relative location of the claimed layers, potential compositions of the respective layers, and their relationships with other structures. *Id.*, ¶ 5. These requirements do not remedy the fact that a POSA, applying Complainant’s vague interpretation of the “layer” terms, would not know which coating(s) of material(s) could be grouped together or subdivided and labeled as “layers” for purposes of the claims. *Id.*

Third, Complainant incorrectly asserts that Respondents’ proposed construction is a “departure from the ordinary meaning” of the term “layer.” Complainant Open. Br. at 7-8.

Respondents' alternative proposed construction is based on dictionary definitions and prior court opinions that reflect the plain-and-ordinary meaning of "layer." Ex. 7 at 9-10 (citing Webster's Dictionary); Ex. 8 at 10 (citing Wiley Electrical and Electronics Engineering Dictionary); *Phillips v. AWH Corp.*, 415 F.3d 1303, 1314 (Fed. Cir. 2005) (en banc) ("general purpose dictionaries may be helpful" for claim construction).

Fourth, Complainant incorrectly argues that Respondents' construction "injects uncertainty into the claims." Complainant Open. Br. at 8. As discussed above, the "separate and defined" portion of Respondents' construction clarifies both that (i) a single coating cannot be subdivided into multiple layers, and (ii) when multiple coatings can be combined into a larger layer. And there is no meaningful dispute that a "layer," in the context of the '546 and '167 patents, is also necessarily a "coating of material," as reflected in Respondents' construction. Banerjee Responsive Decl., ¶ 9-10. The "first liner layer" and "refractory metal layer" in the '167 patent's claims are expressly described as respectively "coating" a sidewall of the contact opening and the first liner layer. *Id.* Similarly, the claims of the '546 patent expressly recite the first liner layer as "conformally coat[ing]" the contact opening and the refractory metal layer as being "over the first intermix region," meaning that the first liner and refractory metal layers are also coatings of materials in the '546 patent. *Id.* All embodiments depict every layer as a coating of material, and a POSA would not have understood the claims of the '167 and '546 patents to cover any "layer" that is not a coating of material. *Id.* Indeed, Complainant provides no example of any "layer" that would not be a coating of material.

For the reasons herein and in Respondents' opening brief, these terms should be construed to mean "a separate and defined [first liner / refractory metal] coating of material" or, alternatively, to be indefinite. Banerjee Responsive Decl., ¶ 13.

B. “intermix region”

Disputed Claim Term	Complainant’s Proposed Construction	Respondents’ Proposed Construction
“intermix region” (’546 patent, claims 1-2, 6, 8-9, and 13; ’167 patent, claims 1, 2, and 6)	Plain and ordinary meaning; not indefinite.	Indefinite

Complainant asserts that an “intermix region” is “a region in which materials from different layers have intermixed.” Complainant Open. Br. at 9. However, neither Complainant nor its expert explains (i) what type of intermixing is required, or (ii) how much intermixing is required for a “region” to qualify as an “intermix region.” Banerjee Responsive Decl., ¶ 14. It is not enough that the claims provide some requirements regarding the relative position of the intermix region or what layer(s) are used to create it. *Id.*; *Berkheimer v. HP Inc.*, 881 F.3d 1360, 1363–64 (Fed. Cir. 2018) (considering term “minimal redundancy,” noting that intrinsic record “lack[ed] objective boundary or specific examples of what constitute[d] ‘minimal’ in the claims,” and holding that the term was therefore indefinite); *id.* at 1364 (“the issue is not *what* must exhibit minimal redundancy, but rather *how much* is minimal”). Similar to *Berkheimer*, the record here fails to provide reasonable certainty regarding **how much** intermixing is required to create an “intermix region.”

In addition, and more generally, there is no evidence regarding what qualifies as an intermix region or how to identify the boundary between an intermix region and other structures. *See* Respondents Open. Br. at 11-14. For example, there is no explanation in the intrinsic record about how uniformly intermixed the composition of an intermix region must be or how intermixing should be measured, further demonstrating indefiniteness. *See S.O.I.TEC Silicon On Insulator Techs., S.A. v. MEMC Elec. Materials, Inc.*, 745 F. Supp. 2d 489, 508–509 (D. Del. 2010) (construing “thermal annealing of the semiconductor material substrate at sufficiently low temperature to substantially limit diffusion of gas” and finding the term indefinite because, *inter*

alia, “diffusion is not a phenomenon that occurs uniformly,” there was “no disclosure how diffusion of free gas from the wafer is to be measured,” and “no indication of how much diffusion is too much” in the record).

Complainant cites examples from the specification of intermix regions. Complainant Open. Br. at 10. But the specification’s non-limiting descriptions do nothing to delineate the boundaries of the *claims* that include this term with reasonable certainty. *Decisioning.com, Inc. v. Federated Dep’t Stores, Inc.*, 527 F.3d 1300, 1314 (Fed. Cir. 2008).

There is clear and convincing evidence that the term “intermix region” is indefinite.

C. “from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide”

Disputed Claim Term	Complainant’s Proposed Construction	Respondents’ Proposed Construction
“from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide” (’235 patent, claims 1 and 10)	Plain and ordinary meaning; not indefinite.	Indefinite

Complainant ignores the issue Respondents identified during the meet-and-confer process: the specification fails to explain *how* to identify “a nickel silicide surface at a silicon interface” or “an opposite surface of the nickel silicide” in an actual product. Because there are multiple methodologies for determining the location of these interfaces/surfaces, and these different methodologies would yield materially different results, this term is indefinite. Ernst Decl., ¶¶ 37-61; *Teva Pharms. USA, Inc. v. Sandoz, Inc.*, 789 F.3d 1335, 1343-45 (Fed. Cir. 2015); *Dow Chem. Co. v. Nova Chems. Corp.*, 803 F.3d 620, 634-35 (Fed. Cir. 2015).

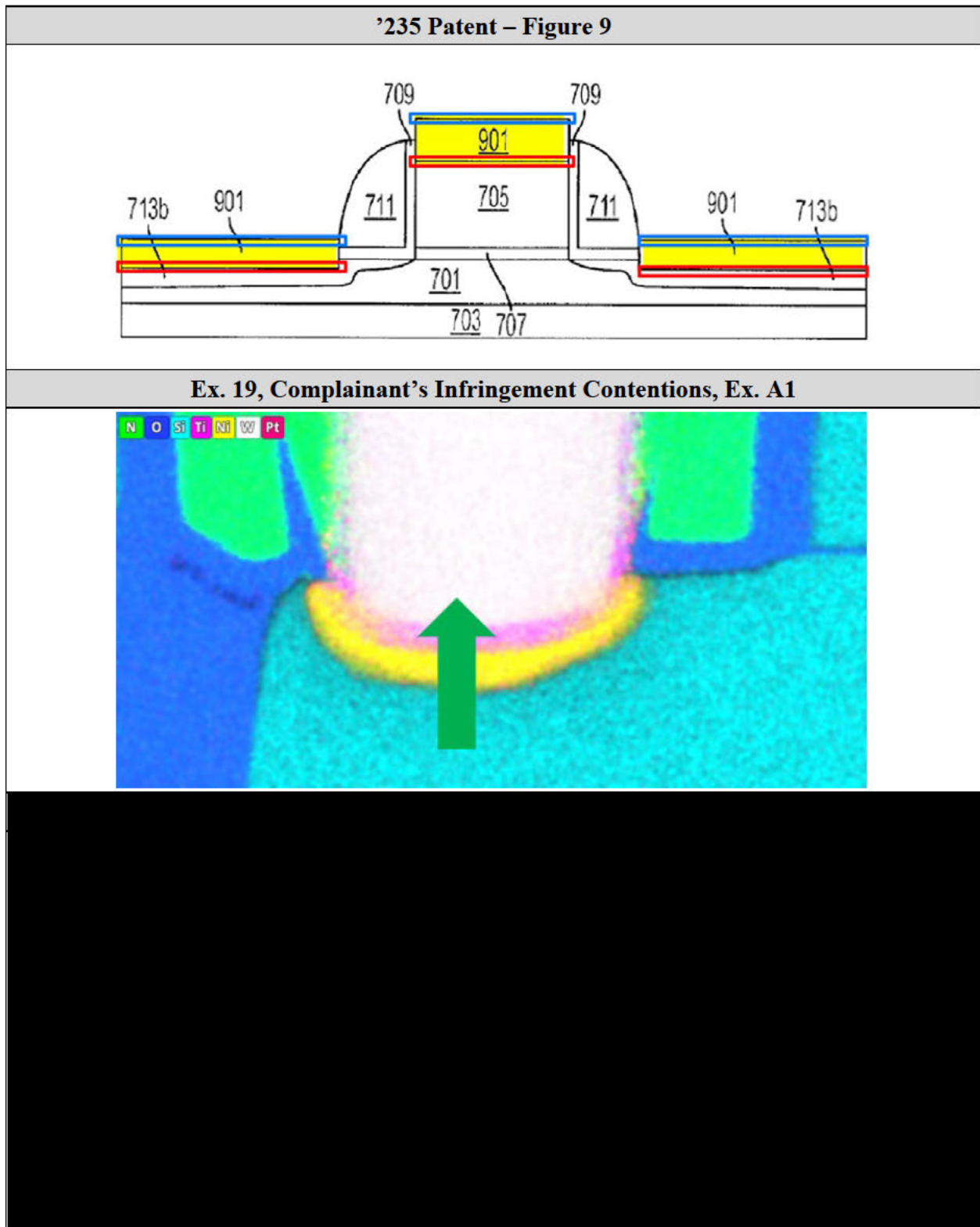
Complainant’s arguments, and Dr. Liu’s conclusory declaration (Liu Decl., ¶¶ 43-48), do not change the indefiniteness of this term.

First, Complainant argues that “[a] POSA would understand that the disputed term identifies the endpoints of a platinum composition gradient, namely, from the bottom of the nickel silicide (a nickel silicide surface ‘at a silicon interface’) to the top of the silicide (‘an opposite surface of the nickel silicide’).” Complainant Open. Br. at 10-11. This, if anything, only highlights the indefiniteness of the claim term. Because the gradient is established by the “endpoints,” the actual location of the “endpoints” (and how to determine where those locations are) becomes critical to understanding the scope of the claims. Because the ’235 patent specification provides no guidance on making this critical determination, the claim is indefinite.

Second, Complainant suggests that because the claim language at issue was added in response to an Examiner’s rejection, and the claims were subsequently allowed by the Examiner, that this supports definiteness. Complainant Open. Br. at 11. Complainant’s suggestion is meritless. The fact that the Examiner suggested adding “endpoints” to overcome prior art does nothing to change that the patent specification fails to provide guidance on how to locate those “endpoints” in an actual product. Indeed, nothing in the Examiner’s rejection, the applicant’s response to that rejection, the Examiner’s amendment, or the Examiner’s subsequent allowance did anything to clarify where the “endpoints” actually are or what methodology should be used to determine their location in an actual product. *See generally* Ex. 47.

Third, Complainant points to generic description in the specification, and a cartoon image from Figure 9 of the patent, as purported support that a POSA would understand this term. Complainant Open. Br. at 11-12. But nothing in the specification or the patent’s cartoon images provides any guidance on how to determine the location of the critical surfaces/interface in an actual product. Indeed, comparing (i) the cartoon image in Figure 9 with (ii) the testing images of actual products that Complainant relies on its contentions only emphasizes the issue. As shown

below, while the cartoon image has straight lines clearly delineating the surfaces, images of actual products have no such lines:



Fourth, Complainant argues that “the specification provides examples of *why* these endpoints [e.g., the location of a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide] matter.” Complainant Open. Br. at 12 (emphasis added). But stating *why* the “endpoints” are important does not provide any clarity to a POSA on *where* the “endpoints” are located or *how* (i.e., what methodology should be used) to determine the “endpoints” location. The lack of guidance in the specification and prosecution history results in indefiniteness.

Complainant apparently wants to rely on a “plain-and-ordinary meaning” construction so that it, and its expert, can arbitrarily choose the location of the claimed interfaces/surfaces to support their position. This is legally improper. *Teva*, 789 F.3d at 1342 (“A party cannot transform into a factual matter the internal coherence and context assessment of the patent simply by having an expert offer an opinion on it. The internal coherence and context assessment of the patent, and whether it conveys claim meaning with reasonable certainty, are questions of law.”). There can be no dispute that the patent specification and prosecution history fail to provide any guidance regarding how (i.e., what methodology should be used) to determine the actual locations of the claimed “opposite surface” and “nickel silicide surface at a silicon interface” in an actual product. Since there are multiple methodologies which lead to different results, the claim term is indefinite. Ernst Decl., ¶¶ 37-61; *see, e.g., Dow*, 803 F.3d at 634–35.

D. “adjacent to” and “the channel is displaced from [a/the] first heavily doped region adjacent to the first edge of the gate”

Disputed Claim Term	Complainant’s Proposed Construction	Respondents’ Proposed Construction
“adjacent to” (’983 patent, claims 1, 5, 13, 17, and 20)	Plain and ordinary meaning; not indefinite.	Indefinite
“the channel is displaced from [a/the] first heavily doped region adjacent to the first edge of the gate” (’983 patent, claims 1, 13, and 20)	Plain and ordinary meaning; not indefinite.	Indefinite

The terms “adjacent to” and “the channel is displaced from [a/the] first heavily doped region adjacent to the first edge of the gate” are indefinite for failing to provide a POSITA reasonable certainty regarding the scope of the claims. Complainant offers nothing to suggest otherwise—no proposed construction, no answer to the myriad unresolved questions of claim scope, no objective standard or guideposts in the intrinsic or extrinsic evidence, nothing to support any reasonable certainty as to claim scope. And yet as Respondents pointed out in their opening brief, Complainant argues in its validity contentions that there is some unidentified distance beyond which two elements are not “adjacent to” each other in the prior art. Respondents Open. Br. at 25-26. Complainant cannot have it both ways—*i.e.*, no reasonably certain scope, but Complainant can draw an arbitrary line in an effort to distinguish invalidating prior art. *See Interval Licensing LLC v. AOL, Inc.*, 766 F.3d 1364, 1371 (Fed. Cir. 2014) (“a term of degree fails to provide sufficient notice of its scope if it depends ‘on the unpredictable vagaries of any one person’s opinion’” (citation omitted)); *In re Walter*, 698 F. App’x. 1022, 1026 (Fed. Cir. 2017) (finding indefinite “a term of degree without any accompanying guidance in the intrinsic record for determining its scope”). The Court should find all claims indefinite.

Complainant and its expert’s inability to articulate any objective standard for the scope of the claims affirms their indefiniteness. Complainant does not even attempt to offer a construction—instead, it yet again hides behind the purported “[p]lain and ordinary meaning.” Complainant does not offer any upper limit, guideline distance, functional standard, or any other objective guideposts for what constitutes “adjacent to” or “displaced from.” Instead, Complainant offers a generic dictionary definition for “adjacent”—“situated near or next”—that offers no objective guidance and merely begs the same question as the claim language itself: what is the scope? Complainant Open. Br. at 13. Complainant marches through the specification evidence, which merely recites the claim language “adjacent to” and provides one example diagram (Figure 1) with *no* guidance on what distances would or would not constitute “adjacent to” or “displaced from” in the Figure 1 device or any other. *Id.* at 13-15. And Complainant points to claim 2 to argue that “displaced from” is definite because the claimed “spacer” defines it, even though that yet again begs the question: what range of distances can the spacer be? Complainant’s declaration of David Liu merely recites the same evidence as Complainant’s brief and adds conclusory opinions that a POSITA would have understood the claim scope with reasonable certainty. Such conclusory expert testimony should be disregarded. *Phillips*, 415 F.3d at 1318 (“conclusory, unsupported assertions by experts as to the definition of a claim term are not useful to a court”).

Complainant’s cases are inapposite, including because they apply the pre-*Nautilus* insolubly ambiguous indefiniteness standard no longer in force. Complainant Open. Br. at 13-15;² *Accentra, Inc. v. Staples, Inc.*, 500 F. App’x 922, 930 (Fed. Cir. 2013) (citing “insolubly

² GF’s remaining cases are irrelevant. *MBO Lab’ys, Inc. v. Becton, Dickinson & Co.*, 474 F.3d 1323, 1333 (Fed. Cir. 2007) (no indefiniteness challenge, and pre-*Nautilus*); *Cox Commc’ns, Inc. v. Sprint Commc’ns Co.*, 838 F.3d 1224, 1232–33 (Fed. Cir. 2016) (pertaining to functional claim language).

ambiguous” standard); *Power-One, Inc. v. Artesyn Techs., Inc.*, 599 F.3d 1343, 1348 (Fed. Cir. 2010) (same); *see also Nautilus*, 572 U.S. at 910 (applying “reasonable certainty” standard).

Even if Complainant’s cases had applied the correct standard, they merely underline the ’983 patent’s deficiencies. As shown in Respondents’ opening brief and the accompanying expert declaration of Dr. Choi, the ’983 patent provides no guidance for the range of distances encompassed by “adjacent to,” instead merely reciting the “adjacent to” language with no restrictions or guideposts on the range of distances. *E.g.*, Choi Decl., ¶¶ 44-53. Further, the technology area of the ’983 patent—laterally diffused metal oxide semiconductor (LDMOS) devices—only exacerbates the lack of reasonable certainty a POSITA would have had. ’983 patent, 1:16-31, 2:45-47; Choi Decl., ¶ 47 (“A POSITA would have particularly lacked any objective guidance in the context of LDMOS devices, since such devices are laterally diffused and therefore can have significant variation in terms of the distances between laterally distributed structures in the device.”). In contrast, in the *Accentra* case, the Court found that the term “near” in the claim language “a pressing area near a front end of the handle” of a stapler patent was not indefinite, because “[s]taplers require leverage on the handle to operate and, as the patent makes clear, users typically apply pressure near the front of the handle, where the lever’s mechanical advantage is the greatest. In this context, we find that the term is sufficiently precise to describe the location of the area that is pressed to discharge staples.” *Accentra*, 500 F. App’x at 930 (emphasis added). Similarly, in the *Power-One* case, the Court found the term “near” was not indefinite because the patent provided “guidance on where [the relevant component] is to be located,” including a “functionality requirement [that] restricts the boundaries of where the regulator can be located in relation to the load it is powering.” *Power-One*, 599 F.3d at 1349. The ’983 patent provides no such guidance for the “adjacent to” term.

The Court should find all claims of the '983 patent invalid based on the indefiniteness of the “adjacent to” and “the channel is displaced from [a/the] first heavily doped region adjacent to the first edge of the gate” claim terms.

E. “the drift isolation region is disposed under the gate”

Disputed Claim Term	Complainant’s Proposed Construction	Respondents’ Proposed Construction
“the drift isolation region is disposed under the gate” ('983 patent, claims 4 and 16)	Plain and ordinary meaning	“the drift isolation region is positioned at least in part under the gate”

Respondents understand that Complainant has dropped its position and adopted Respondents’ proposed construction.

F. “the channel having a channel length that is controlled by ion implantation”

Disputed Claim Term	Complainant’s Proposed Construction	Respondents’ Proposed Construction
“the channel having a channel length that is controlled by ion implantation” ('983 patent, claims 7 and 18)	Plain and ordinary meaning	“the channel having a channel length that is formed by implanting ions into the substrate to form a channel well”

The Court should adopt Respondents’ construction for “the channel having a channel length that is controlled by ion implantation,” since it clarifies the plain and ordinary meaning. That clarification is essential given that Complainant continues to suggest it has some particular (and unsupported) understanding of the claim language while refusing to articulate what that understanding is. Respondents’ construction prevents Complainant from continuing to hide the ball from Respondents and the Court.

Complainant now suggests there is some undefined difference between “controlled” and “formed.” Complainant Open. Br. at 17. Complainant argues that “while components (such as

the ‘channel’) are ‘formed,’ dimensions of such components (such as the ‘channel length’) are ‘controlled.’” *Id.* But Complainant is simply wrong. Contrary to Complainant’s characterization, the specification *does* disclose “forming” channel lengths. *E.g.*, ’983 patent, 1:34-35 (“This makes **forming short channel lengths** to achieve low R_{Dson} difficult if not impossible.” (emphasis added)), 1:39-41 (“devices formed with short channel lengths”).

Complainant’s argument also defies logic, because the dimensions of the channel length result from how components like the channel and channel well are formed. *Id.*, 1:31-33 (“the HV channel well, which defines the channel length”), 5:53-56 (“The channel edge of the channel well **135** and the first edge **179a** of the gate on the source side define a channel of the HV transistor having an effective channel length L .”), 6:1-4 (“With the channel well being self-aligned to the gate edge of the implant, process overlay issues are avoided. Very short effective channel lengths with very low R_{Dson} can be achieved[.]”). Indeed, the specification discloses the channel length being “controlled by channel or body implant conditions”—*i.e.*, the implantation of ions in the channel well. *Id.*, 5:62-65. Tower’s construction clarifies the relationship between implanting ions to form a channel well and forming a channel length, while Complainant’s purported “plain and ordinary meaning” appears to deny that relationship exists, despite the specification evidence.

Instead of allowing Complainant to pursue an unexplained interpretation of the claim language directly contradicted by the specification, the Court should adopt Respondents’ construction.

G. “edge of the gate”

Disputed Claim Term	Complainant’s Proposed Construction	Respondents’ Proposed Construction
“edge of the gate” (’983 patent, claims 1, 2, 6, 13, 14, 15, and 20)	Plain and ordinary meaning	“edge of the gate electrode over the gate dielectric”

Respondents’ construction clarifies the plain and ordinary meaning of “edge of the gate” in the context of the intrinsic evidence. The claims recite that the gate includes “a gate electrode over a gate dielectric.” ’983 patent, 13:18-19, 14:6-7, 14:44-45. Thus, the edge of the “gate” according to the claims must be the edge of where there is a “gate electrode over a gate dielectric.” Complainant disagrees, but the *only* evidence it cites in the entire “edge of the gate” section of its opening brief is where the specification discloses “a gate with first and second edges **179a-b** or sidewalls.” Complainant Open. Br. at 17-18 (citing ’983 patent, 4:46-51). Figure 1 shows that these “first and second edges **179a-b** or sidewalls” are the edge of the gate electrode over the dielectric. ’983 patent, Fig. 1. Accordingly, Complainant continues to fail to articulate its understanding of the “scope of the edge of the gate” or any disagreement with Respondents’ construction, just as it did during the meet-and-confer process. For example, during the meet-and-confer process, Respondents presented Complainant with the following diagram and asked which of the arrows are pointing to a potential “edge of the gate.”

Complainant argues that Respondents improperly equate the “gate” with the “gate electrode,” when the gate can include other elements. Complainant Open. Br. at 17. But Complainant is wrong. Respondents’ construction is compatible with the gate including other elements. Regardless of what other elements the gate may include, its edges are the edges of the

gate electrode over the gate dielectric as the claims recite. '983 patent, 13:18-19 (“the gate including a gate electrode over a gate dielectric”), 14:6-7 (same), 14:44-45 (same).

Complainant also argues that “[t]here is no disclaimer or lexicography to support limiting the ‘edge of the gate’ to only portions ‘over the gate dielectric[.]’” Complainant Open. Br. at 17-18. But there is unambiguous claim language defining a gate at minimum as “a gate electrode over a gate dielectric[.]” '983 patent, 13:18-19, 14:6-7, 14:44-45; *Phillips*, 415 F.3d at 1312 (“we look to the words of the claims themselves ... to define the scope of the patented invention” (citation omitted)). And as Dr. Choi opined in his declaration, this claim language would have been consistent with a POSITA’s understanding based on knowledge of electronics that a “gate” only exists in a MOSFET where there is a gate electrode over a gate dielectric. Choi Decl, ¶¶ 55-56.

The Court should adopt Respondents’ construction and prevent Complainant from hiding its understanding of the “edge of the gate” claim language behind its “plain and ordinary meaning” proposal.

III. CONCLUSION

For the reasons above and in Respondents’ opening brief, there is clear and convincing evidence that all claims of the '546, '167, '235, and '983 patents are invalid for indefiniteness. If the CALJ is not inclined to find indefiniteness on this record, Respondents respectfully request that the CALJ defer consideration of indefiniteness until after the close of fact and expert discovery. In addition and/or the alternative, Respondents respectfully request adoption of their proposed constructions for the reasons above.

CERTIFICATE OF WORD COUNT

Pursuant to Ground Rules 1.7 and 7.2, the undersigned hereby certifies that the foregoing contains 4,982 words, including footnotes but excluding words in the caption, tables of contents/authorities/exhibits, figures, signature block, this Certificate of Word Count, and the Certificate of Service. Furthermore, this brief complies with the 5,000-word limit for responsive claim construction briefs in Ground Rule 7.2. In preparing this certificate, the undersigned relied on the word count of the Microsoft Word program.

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